

Sodium, Water, Potassium, and Divalent Cation Disorders

An original clinical reasoning chapter for electrolyte disorders, designed for study and future graph-RAG extraction. Uploaded renal source material and Anki cards were used as coverage signals only; wording, diagrams, cases, and tables are original.

Clinical frame

Electrolyte questions are rarely about an isolated laboratory value. The lab value is the surface finding; the clinical task is to decide whether the patient has dangerous tonicity, dangerous membrane instability, dangerous volume status, or a chronic adaptation that will be harmed by fast correction.

Learning Objectives

- Classify hyponatremia by serum osmolality, volume status, urine osmolality, and urine sodium rather than by serum sodium alone.
- Treat symptomatic hyponatremia with hypertonic saline while avoiding osmotic demyelination from overly rapid correction.
- Differentiate water loss, sodium gain, diabetes insipidus, and osmotic diuresis in hypernatremia.
- Recognize ECG-dangerous potassium disorders and separate stabilization, shifting, and potassium removal therapies.
- Use PTH, phosphate, magnesium, kidney function, medication history, and malignancy clues to classify calcium disorders.
- Connect phosphate and magnesium abnormalities to renal failure, refeeding, alcohol use disorder, torsades, and refractory hypokalemia or hypocalcemia.

Source coverage anchor

The uploaded renal chapter emphasized volume compartments, urine output, normal saline, lactated Ringer solution, hypervolemia, sodium homeostasis, hyponatremia, hypernatremia, calcium metabolism, and hypocalcemia. It specifically flagged the 60-40-20 fluid compartment rule, urine sodium patterns in hyponatremia, hypertonic saline for severe symptomatic hyponatremia, cautious correction rates, diabetes insipidus as a cause of hypernatremia, and PTH/calcitonin/vitamin D control of calcium. This chapter expands those anchors into a full electrolyte framework.

Original diagram: electrolyte logic along the nephron



1. The Four Questions in Every Electrolyte Case

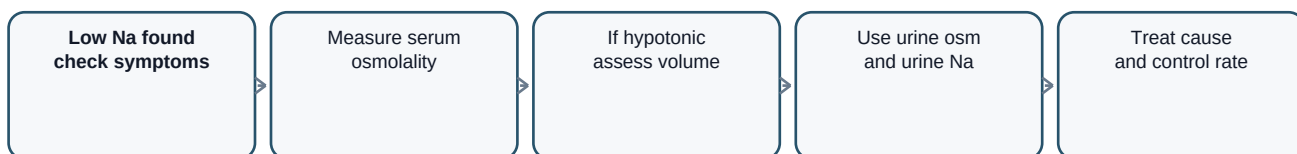
The safest approach begins by asking four questions before prescribing a correction. First, is there immediate electrical or neurologic danger? Severe hyperkalemia with QRS widening, severe hypocalcemia with seizures or laryngospasm, severe hypercalcemia with coma, and acute hyponatremic encephalopathy are emergencies. Second, is the disorder acute or chronic? Brain cells adapt to chronic tonicity changes over roughly 48 hours, so chronic sodium abnormalities must be corrected slowly. Third, is the abnormality a problem of amount, distribution, or measurement? Pseudohyponatremia and hyperglycemia-associated translocational hyponatremia are not true hypotonic water excess. Finally, is the kidney responding appropriately? Urine osmolality and urine electrolytes reveal whether ADH, aldosterone, diuretics, renal failure, or extrarenal loss is driving the pattern.

Question	Why it matters	High-yield data
Is this immediately dangerous?	Treat the patient before completing the whole diagnostic algorithm.	ECG, seizures, coma, severe weakness, arrhythmia, Ca >14 mg/dL, K >=6.5 mEq/L
Acute or chronic?	Rapid correction of chronic hyponatremia can cause osmotic demyelination; rapid correction of chronic hypernatremia can cause cerebral edema.	Symptom timing, prior labs, alcoholism/malnutrition/liver disease risk
True or artifact/translocation?	Low sodium with normal/high tonicity has different management from true hypotonic hyponatremia.	Measured serum osmolality, glucose, triglycerides, protein
Is the kidney behaving appropriately?	Urine studies often reveal the mechanism.	Urine osmolality, urine Na, urine K, urine Cl, FE indices when relevant

2. Sodium and Water: Tonicity Before Volume

Serum sodium is best understood as a tonicity signal, not as a total-body sodium inventory. Hyponatremia usually means too much water relative to effective osmoles; hypernatremia usually means too little water relative to effective osmoles. Volume status still matters because hypovolemia, heart failure, cirrhosis, kidney failure, adrenal insufficiency, and SIADH each produce different neurohormonal signals and different urine patterns. A patient may have low total-body sodium and hyponatremia, normal total-body sodium and hyponatremia, or high total-body sodium and hyponatremia depending on whether the primary process is salt loss, ADH-mediated water retention, or edema-forming sodium retention with even more water retention.

Diagnostic sequence for low sodium



Hyponatremia

Hyponatremia is typically defined as serum sodium below 135 mEq/L. Symptoms often correlate more with acuity than with the absolute number. A rapid fall from 140 to 124 may cause vomiting, headache, confusion, seizure, or coma, while a chronic sodium of 124 may be relatively quiet. The dangerous organ is the brain: water shifts into brain cells during acute hypotonicity, producing cerebral edema. With chronicity, the brain extrudes osmoles and becomes vulnerable to dehydration injury if the serum sodium is raised too quickly.

Category	Mechanism	Urine pattern	Common examples	Initial management logic
Hypovolemic hypotonic hyponatremia	Salt and water lost, but salt loss dominates; ADH rises because effective arterial volume is low.	Urine osm usually >100 ; urine Na $<20-30$ if extrarenal loss, often >30 if renal salt loss/diuretic/adrenal issue.	Vomiting, diarrhea, burns, pancreatitis, third spacing, diuretics, mineralocorticoid deficiency.	Restore effective volume with isotonic crystalloid; sodium often corrects as ADH turns off.
Euvolemic hypotonic hyponatremia	Water retained without obvious edema; total body sodium is near normal.	Urine osm >100 in SIADH; urine Na often >30 ; low uric acid may support SIADH.	SIADH from pulmonary/CNS disease, medications, malignancy; adrenal insufficiency; hypothyroidism; low solute intake; primary polydipsia.	Fluid restriction for many stable cases; treat endocrine causes; hypertonic saline if neurologically symptomatic.
Hypervolemic hypotonic hyponatremia	Total body sodium is high, but water retention is even greater because effective arterial volume is low.	Urine Na often low unless renal failure or diuretics are present.	Heart failure, cirrhosis, nephrotic syndrome, advanced kidney failure.	Restrict free water and sodium; diurese if congested; treat underlying organ failure.
Isotonic/pseudohyponatremia	Lab artifact from marked lipids/proteins with normal plasma tonicity.	Not classified by hypotonic algorithm.	Severe hypertriglyceridemia, paraproteinemia.	Treat the underlying condition; do not give hypertonic saline for artifact.
Hypertonic/translocational hyponatremia	Effective extracellular osmoles pull water out of cells and dilute sodium.	Depends on cause.	Hyperglycemia, mannitol, radiocontrast in selected settings.	Correct effective osmole; estimate glucose-corrected sodium.

Correction principle

For severe symptomatic hyponatremia, the goal is not to normalize sodium rapidly. The immediate goal is a small rise, often 4 to 6 mEq/L, enough to reduce cerebral edema and stop seizures. Many guideline discussions warn against exceeding about 8 mEq/L in 24 hours in high-risk patients, and overcorrection should prompt active relowering strategies such as desmopressin and free water under expert guidance.

SIADH as a script

SIADH is an ADH-excess physiology: the collecting duct reabsorbs water even when plasma is already hypotonic. The typical script is hypotonic hyponatremia, inappropriately concentrated urine, continued urinary sodium excretion, clinical euvolemia, low serum uric acid in many cases, and absence of adrenal, thyroid, kidney, or diuretic explanations. Causes cluster into pulmonary disease, CNS disease, postoperative pain/nausea/stress, medications such as SSRIs or carbamazepine, and ectopic ADH production such as small cell lung cancer. Stable SIADH is often treated first with fluid restriction, increased solute intake, and removal of the trigger; vaptans are specialized options and are not a reflex answer for every patient.

Hypernatremia

Hypernatremia is generally serum sodium above 145 mEq/L and almost always represents water deficit, impaired access to water, or less commonly sodium gain. Because thirst is a powerful defense, sustained hypernatremia is a clue to altered mental status, intubation, infancy, older age, limited mobility, lack of access to water, hypothalamic disease, diabetes insipidus, or large renal/GI water losses. The clinical danger is cellular dehydration, especially in the brain. Acute hypernatremia can cause weakness, irritability, seizures, or intracranial hemorrhage; overly rapid treatment of chronic hypernatremia can cause cerebral edema.

Volume phenotype	Core mechanism	Urine clue	Examples	Treatment sequence
Hypovolemic hyponatremia	Loss of water and sodium, but water loss exceeds sodium loss.	Appropriately concentrated urine if extrarenal loss; high urine sodium if renal loss/diuretics/osmotic diuresis.	Diarrhea, osmotic diuresis, diuretics, burns, fever, sweating.	If unstable, restore perfusion first with isotonic fluid; then replace free water deficit.
Euvolemic hyponatremia	Predominant water loss with little sodium change.	Very dilute urine suggests diabetes insipidus; concentrated urine suggests extrarenal water loss or inadequate intake.	Central DI, nephrogenic DI, insensible losses, limited water intake.	Free water by mouth/enteral route or D5W; DDAVP if central DI.
Hypervolemic hyponatremia	Sodium gain exceeds water gain.	Often iatrogenic context.	Hypertonic bicarbonate/saline, salt ingestion, mineralocorticoid excess, TPN errors.	Stop sodium source; use loop diuretics plus free water if kidneys work; dialysis if severe kidney failure or overload.

Free water deficit

A practical estimate is: water deficit = TBW x [(current Na / desired Na) - 1]. Use roughly 0.6 x weight in kg for younger men, 0.5 for women or older men, and lower estimates in frail older adults. The formula is a starting point; ongoing urine, stool, fever, and ventilator losses must be added.

Diabetes Insipidus and Primary Polydipsia

Polyuria syndromes are best recognized by urine volume and urine tonicity. Diabetes insipidus produces excessive dilute urine despite hypertonicity because ADH is absent or ineffective. Central DI is impaired vasopressin release and responds to desmopressin. Nephrogenic DI is renal resistance to ADH; classic causes include lithium, hypercalcemia, hypokalemia, kidney disease, and inherited collecting duct defects. Primary polydipsia can also produce very dilute urine, but the driver is excess intake rather than failed ADH signaling. In uncertain cases, water deprivation with desmopressin response or copeptin-based protocols may be used in specialized settings.

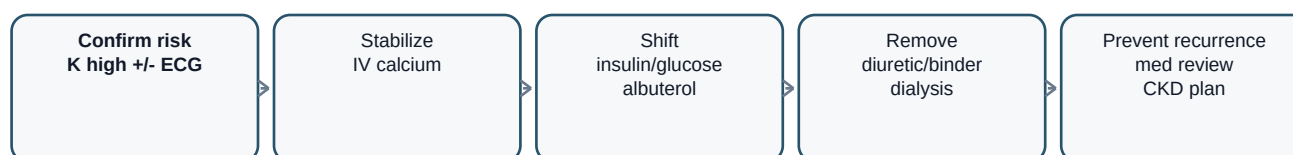
Disorder	Serum	Urine	Response to desmopressin	Treatment anchors
Central DI	High or high-normal Na/osmolality if intake cannot keep up	Dilute, often urine osm <300 mOsm/kg	Urine osmolality rises substantially	Desmopressin; ensure water access; avoid overreplacement causing hyponatremia
Nephrogenic DI	High or high-normal Na/osmolality	Dilute despite ADH signal	Little response	Remove cause; low-solute diet; thiazide; amiloride for lithium-associated disease; sometimes NSAID specialist use
Primary polydipsia	Low-normal sodium possible	Dilute urine due to excess intake	Limited diagnostic value without protocol	Behavioral/psychiatric treatment; avoid rapid water restriction in high-risk contexts

3. Potassium: Membrane Stability First

Potassium disorders are dangerous because extracellular potassium strongly controls resting membrane potential. Serum potassium is also an imperfect proxy for total-body potassium: acid-base shifts, insulin, beta-adrenergic tone, tissue breakdown, and osmolality can move potassium between compartments without changing total stores. The first question is whether the myocardium or skeletal muscle is unstable. When hyperkalemia causes ECG changes, calcium protects the membrane but does not remove potassium. Insulin with glucose and beta-agonists shift potassium into cells; diuretics, binders, and dialysis remove potassium from the body.

Disorder	Key causes	ECG/clinical clues	Treatment logic
Hypokalemia	GI loss, loop/thiazide diuretics, hyperaldosteronism, alkalosis, insulin/beta-agonists, poor intake, hypomagnesemia.	Weakness, cramps, ileus, flattened T waves, ST depression, U waves, ventricular arrhythmias, increased digoxin toxicity.	Replace K, correct Mg, identify renal vs extrarenal losses with urine K/acid-base/BP pattern.
Hyperkalemia	Kidney failure, ACEi/ARB, mineralocorticoid deficiency/resistance, potassium-sparing diuretics, trimethoprim, heparin, acidosis, tissue breakdown, insulin deficiency, pseudohyperkalemia.	Peaked T waves, PR prolongation, P-wave flattening, QRS widening, sine wave, weakness/paralysis; ECG can be deceptively normal.	If ECG changes or severe level: IV calcium, shift with insulin/glucose plus albuterol, remove K, stop contributors, consider dialysis.

Emergency hyperkalemia sequence



Hypokalemia scripts

The most efficient way to classify hypokalemia is to pair blood pressure, acid-base status, and urinary potassium. Low potassium with metabolic alkalosis and low urine chloride suggests vomiting or remote diuretic use; low potassium with metabolic alkalosis and high urine chloride suggests ongoing diuretics, Bartter/Gitelman physiology, or mineralocorticoid excess. Hypertension points toward aldosterone or apparent mineralocorticoid excess. Non-anion gap metabolic acidosis with low potassium suggests diarrhea or renal tubular acidosis. Hypomagnesemia must be corrected because magnesium depletion removes the brake on renal potassium secretion, making hypokalemia refractory.

Pattern	Likely mechanism	Examples
Low K + metabolic alkalosis + low urine Cl	Chloride-responsive alkalosis from gastric acid loss	Vomiting, nasogastric suction
Low K + metabolic alkalosis + high urine Cl + normal/low BP	Renal salt wasting	Loop/thiazide diuretics, Bartter, Gitelman
Low K + metabolic alkalosis + hypertension	Mineralocorticoid effect	Primary aldosteronism, Cushing, licorice, Liddle
Low K + non-anion gap acidosis	Bicarbonate loss or impaired acid secretion	Diarrhea, proximal/distal RTA

4. Calcium: PTH First, Then Phosphate and Context

Calcium disorders require attention to albumin, pH, phosphate, magnesium, kidney function, and symptoms. Total calcium is partly protein-bound; ionized calcium is the active fraction. Low albumin can make total calcium look low even when ionized calcium is normal. Alkalosis increases albumin binding and can cause symptoms of low ionized calcium even when total calcium is normal. The first diagnostic branch in hypercalcemia is whether PTH is appropriately suppressed. If PTH is high or inappropriately normal, think primary hyperparathyroidism, tertiary hyperparathyroidism, lithium effect, or familial hypocalciuric hypercalcemia. If PTH is low, think malignancy, vitamin D excess, granulomatous disease, immobilization, thyrotoxicosis, adrenal insufficiency, milk-alkali syndrome, or medications.

Hormone/axis	Calcium effect	Phosphate effect	Mechanism
PTH	Raises Ca	Lowers phosphate	Bone resorption, distal tubule Ca reabsorption, proximal phosphate wasting, vitamin D activation
Calcitriol	Raises Ca	Raises phosphate	Increases gut absorption of both Ca and phosphate
Calcitonin	Lowers Ca modestly	Variable/minor	Inhibits osteoclast activity; useful short-term in severe hypercalcemia
FGF23	Indirectly lowers Ca tendency	Lowers phosphate	Promotes renal phosphate wasting and reduces calcitriol activation

Hypercalcemia

Hypercalcemia often produces dehydration because it impairs renal concentrating ability and causes nephrogenic DI-like polyuria. The classic symptom cluster includes stones, bones, abdominal groans, psychiatric overtones, constipation, shortened QT interval, and kidney injury. Severe or symptomatic hypercalcemia requires volume repletion first unless contraindicated, often followed by calcitonin for rapid temporary effect and an antiresorptive agent such as a bisphosphonate or denosumab depending on kidney function and cause. Dialysis is reserved for selected life-threatening cases with kidney failure, severe volume overload, or refractory calcium.

PTH result	Phosphate pattern	Likely diagnoses	Next tests/notes
High or inappropriately normal	Low/normal phosphate	Primary hyperparathyroidism, lithium, tertiary HPT, FHH	24-hour urine calcium or Ca/Cr clearance ratio helps distinguish FHH from primary HPT
Low	Low phosphate	PTHrP malignancy	Squamous cell tumors, renal/bladder/ovarian cancers; check PTHrP when suspected
Low	High/normal phosphate	Vitamin D excess, granulomatous disease, lymphoma, immobilization, renal failure contexts	Check 25-OH vitamin D, 1,25-OH ₂ vitamin D, SPEP/UPEP when appropriate

Hypocalcemia

Hypocalcemia presents with neuromuscular excitability: perioral numbness, paresthesias, cramps, carpopedal spasm, hyperreflexia, tetany, seizures, laryngospasm, and prolonged QT. Etiologies include hypoparathyroidism after neck surgery, autoimmune or genetic hypoparathyroidism, vitamin D deficiency, advanced kidney disease, pancreatitis, sepsis, massive transfusion with citrate, tumor lysis, rhabdomyolysis, hungry bone syndrome, and hypomagnesemia. Magnesium is not optional: severe hypomagnesemia impairs PTH secretion and induces PTH resistance, causing hypocalcemia that will not correct until magnesium is repleted.

PTH	Phosphate	Magnesium	Interpretation
Low	High	Normal/low	Hypoparathyroidism or severe hypomagnesemia suppressing PTH
High	High	Variable	CKD with phosphate retention and reduced calcitriol
High	Low	Variable	Vitamin D deficiency or malabsorption
High	High/variable	Low	Hypomagnesemia causing PTH resistance plus low secretion

5. Phosphate and Magnesium

Phosphate is an intracellular and bone-buffered ion that becomes clinically prominent in kidney disease, refeeding, tumor lysis, rhabdomyolysis, alcoholism, respiratory alkalosis, and endocrine disease. Hypophosphatemia can cause weakness, respiratory failure, hemolysis, rhabdomyolysis, impaired myocardial contractility, and neurologic dysfunction. Hyperphosphatemia can lower calcium through calcium-phosphate precipitation and is common in advanced CKD, tumor lysis, rhabdomyolysis, and hypoparathyroidism. Management depends on severity, symptoms, kidney function, and whether the disorder is a shift, loss, or load problem.

Abnormality	Mechanisms	Clinical anchors	Management anchors
Hypophosphatemia	Refeeding/insulin shift, respiratory alkalosis, alcohol use disorder, poor intake, phosphate binders, proximal tubule wasting, vitamin D deficiency	Weakness, respiratory failure, rhabdomyolysis, hemolysis, encephalopathy	Oral replacement if mild; IV if severe/symptomatic; watch calcium and kidney function
Hyperphosphatemia	CKD retention, tumor lysis, rhabdomyolysis, hemolysis, phosphate enemas, hypoparathyroidism	Hypocalcemia, tetany, vascular/soft-tissue calcification in CKD	Dietary restriction, binders, treat cell lysis, dialysis when severe with kidney failure
Hypomagnesemia	GI loss, alcohol use disorder, PPIs, aminoglycosides, cisplatin, amphotericin, diuretics, refeeding	Tremor, seizures, torsades, refractory hypokalemia, hypocalcemia	Replete magnesium; IV magnesium for torsades/seizures/severe deficiency
Hypermagnesemia	Kidney failure plus magnesium load	Hyporeflexia, weakness, hypotension, bradycardia, respiratory depression	Stop magnesium; IV calcium for severe symptoms; fluids/diuretics or dialysis

Magnesium bridge

Hypomagnesemia is the hidden link between three testable findings: torsades de pointes, refractory hypokalemia, and hypocalcemia. In a patient with alcohol use disorder, diarrhea, PPIs, or diuretics, do not chase potassium or calcium without checking and correcting magnesium.

6. ECG and Emergency Snapshot

Electrolyte	ECG or emergency clue	Immediate action
Hyperkalemia	Peaked T waves, PR prolongation, QRS widening, sine-wave risk	IV calcium if ECG changes/severe; insulin-glucose and albuterol; remove potassium
Hypokalemia	Flattened T, ST depression, U waves, ventricular ectopy	K replacement; telemetry if severe; correct Mg
Hypocalcemia	QT prolongation, tetany, seizures	IV calcium gluconate if symptomatic/severe; Mg and vitamin D/PTH evaluation
Hypercalcemia	Short QT, dehydration, AKI, altered mental status	Isotonic volume repletion, calcitonin, antiresorptive therapy
Hypomagnesemia	Torsades, seizures, refractory K/Ca problems	IV magnesium for torsades or severe symptoms

Common Pitfalls

- Treating a sodium number without checking serum osmolality. Pseudohyponatremia and hyperglycemic translocation are not managed like true hypotonic hyponatremia.
- Assuming edema means intravascular volume excess. Heart failure and cirrhosis can produce edema while baroreceptors sense low effective arterial volume.
- Correcting chronic hyponatremia too quickly after volume repletion. Hypovolemic patients may suddenly water-diurese once ADH shuts off.
- Forgetting that IV calcium stabilizes hyperkalemic myocardium but does not lower total body potassium.
- Replacing potassium without magnesium in refractory hypokalemia.
- Using total calcium alone in hypoalbuminemia or alkalosis when ionized calcium would better match symptoms.
- Missing adrenal insufficiency in a patient with hyponatremia, hyperkalemia, hypotension, eosinophilia, weight loss, or hyperpigmentation.
- Giving desmopressin for suspected DI without monitoring sodium and water intake; over-treatment can convert polyuria into dangerous hyponatremia.

Practice Questions

Q1. A 73-year-old man with small cell lung cancer has Na 119 mEq/L, serum osm 255 mOsm/kg, urine osm 520 mOsm/kg, urine Na 64 mEq/L, normal thyroid/adrenal testing, and no edema. He is confused but not seizing. What is the mechanism?

Answer: Inappropriate ADH activity causing water retention despite hypotonic plasma. The urine is too concentrated and sodium excretion continues despite hyponatremia, supporting SIADH physiology.

Q2. A woman with vomiting has Na 126, metabolic alkalosis, low urine chloride, and orthostatic symptoms. What treatment addresses both sodium and alkalosis?

Answer: Isotonic saline with potassium repletion. Chloride-responsive alkalosis and hypovolemic ADH activation improve when effective volume and chloride are restored.

Q3. A CKD patient has K 7.0 mEq/L with wide QRS complexes. What are the first three treatment categories?

Answer: Stabilize the myocardium with IV calcium, shift potassium intracellularly with insulin/glucose and beta-agonist, and remove potassium with dialysis or other elimination strategies depending on severity and kidney function.

Q4. A patient with alcohol use disorder has K 2.8, Mg 1.0, and persistent hypokalemia despite potassium replacement. Why?

Answer: Low magnesium increases renal potassium wasting and makes hypokalemia refractory. Magnesium must be repleted to retain potassium.

Q5. A patient with hypercalcemia has low PTH and elevated PTHrP. What etiologic class is most likely?

Answer: Humoral hypercalcemia of malignancy, classically associated with squamous cell tumors and other solid tumors.

Q6. A lithium-treated patient has polyuria, hypernatremia, and dilute urine that does not concentrate after desmopressin. What disorder is present?

Answer: Nephrogenic diabetes insipidus from renal resistance to ADH. Lithium is a classic cause; amiloride and low-solute strategies may be used in selected patients.

Graph-RAG Extraction Map

The following map is intentionally explicit for future graph construction. Nodes should preserve numeric anchors, cause-effect links, diagnostic branchpoints, and treatment-action relationships.

Node	Class	Key facts	Suggested edges
Hyponatremia	electrolyte disorder; sodium-water	Na <135; classify by osmolality and volume; severe symptoms reflect cerebral edema	caused_by SIADH; caused_by hypovolemia; treated_by hypertonic saline when severe symptomatic
SIADH	euvolemic hypotonic hyponatremia	Urine osm inappropriately high; urine Na often >30; causes include pulmonary/CNS disease, drugs, small cell lung cancer	causes hyponatremia; diagnosed_by urine osm/urine sodium; treated_by fluid restriction
Hypernatremia	electrolyte disorder; water deficit	Na >145; sustained cases imply impaired thirst/access or water loss	caused_by diabetes insipidus; treated_by free water; complication cerebral edema if overcorrected
Central diabetes insipidus	ADH deficiency	Polyuria, dilute urine, response to desmopressin	treated_by desmopressin; differentiated_from nephrogenic DI
Nephrogenic diabetes insipidus	ADH resistance	Lithium, hypercalcemia, hypokalemia; poor response to desmopressin	caused_by lithium; treated_by thiazide/amiloride in selected cases
Hyperkalemia	potassium disorder; cardiac emergency	Peaked T, QRS widening; risk higher with CKD/RAAS blockade	treated_by IV calcium; treated_by insulin/glucose; treated_by dialysis
Hypokalemia	potassium disorder	Weakness, ileus, U waves; worsens digoxin toxicity	caused_by diuretics; caused_by vomiting; worsened_by hypomagnesemia
Hypercalcemia	divalent cation disorder	Stones/bones/groans/psychiatric; PTH-first algorithm; dehydration common	caused_by primary hyperparathyroidism; caused_by malignancy; treated_by saline/calcitonin/bisphosphonate
Hypocalcemia	divalent cation disorder	Tetany, seizures, QT prolongation; evaluate PTH/phosphate/Mg	caused_by hypoparathyroidism; caused_by vitamin D deficiency; caused_by hypomagnesemia
Hypomagnesemia	magnesium disorder	Torsades; refractory hypokalemia; hypocalcemia	causes hypokalemia; causes hypocalcemia; treated_by magnesium repletion

Selected Accuracy References

1. Society for Endocrinology. Emergency management of severe and moderately severely symptomatic hyponatraemia in adult patients, 2022.
2. Sterns RH. Treatment guidelines for hyponatremia: stay the course. Clinical Journal of the American Society of Nephrology, 2024.
3. KDIGO. Acute hyperkalemia in the emergency department: conference report. Kidney International Reports, 2020.
4. Endocrine Society / Society for Endocrinology patient and clinical guidance on arginine vasopressin deficiency/diabetes insipidus.
5. KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease, used for CKD/electrolyte context.